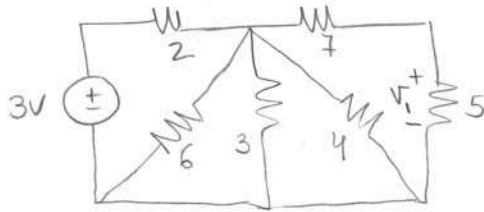
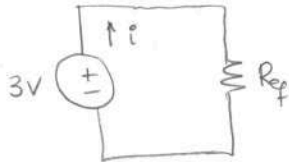


QUESTION 1a



i)

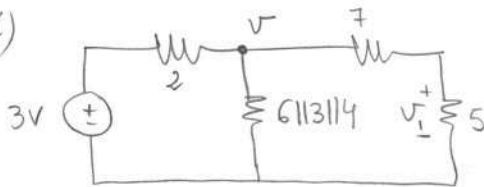


$$R_{eq} = [(5+7) \parallel 4 \parallel 3 \parallel 6] + 2 = 3.2 \Omega$$

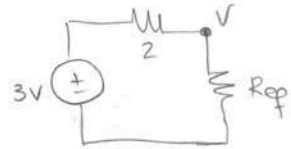
$$i = \frac{3V}{R_{eq}} = \frac{3}{3.2} = 0.938 A$$

$$P_{vs} = -V \times i = -3 \times 0.938 = -2.81 W$$

ii)



$\Rightarrow$



* NOTE: Source transformation + current divider + Ohm's law is straightforward too.

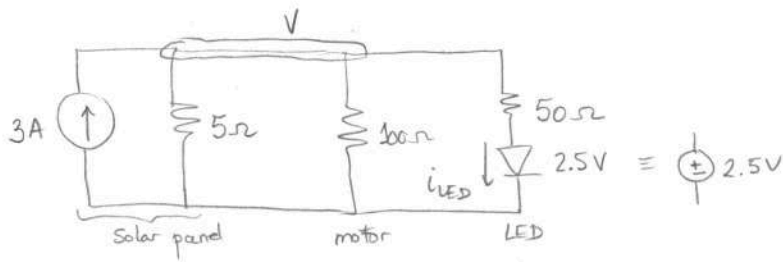
$$R_{eq1} = 6 \parallel 3 \parallel 4 \parallel (7+5) = 1.2 \Omega$$

$$V = 3 \frac{R_{eq1}}{2 + R_{eq1}} = 3 \times \frac{1.2}{3.2} = 1.125 V$$

voltage divider

$$V_1 = V \frac{5}{5+7} = 1.125 \times \frac{5}{12} = 0.469 V$$

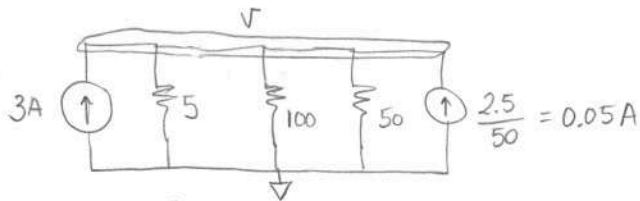
• QUESTION 1b



* Other options:

- Source transformation with $5 \parallel 100$ and KVL
- Mesh analysis
- Nodal analysis

i)



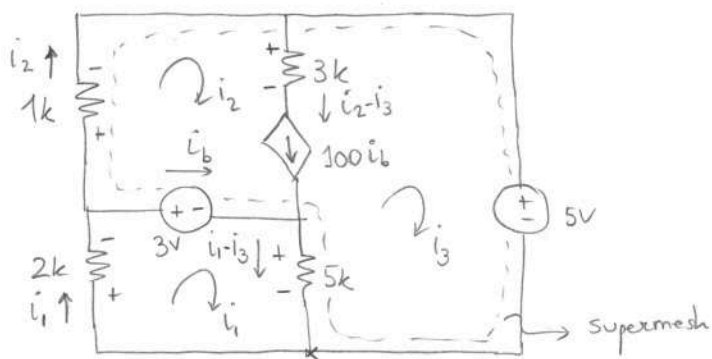
$$3.05 \text{ A} \uparrow \parallel 5 \parallel 100 \parallel 50 \Rightarrow V = 3.05 \times (5 \parallel 100 \parallel 50) = 3.05 \times 4.348 = 13.26 \text{ V}$$

KVL on right mesh: $-V + 50i_{LED} + 2.5 = 0$

$$i_{LED} = \frac{13.26 - 2.5}{50} = 0.215 \text{ A}$$

ii) $P_{50\Omega} = i_{LED}^2 \times R = 0.215^2 \times 50 = 2.31 \text{ W}$

• QUESTION 2a



KVL @ mesh 1: $2000 i_1 + 3 + 5000(i_1 - i_3) = 0$

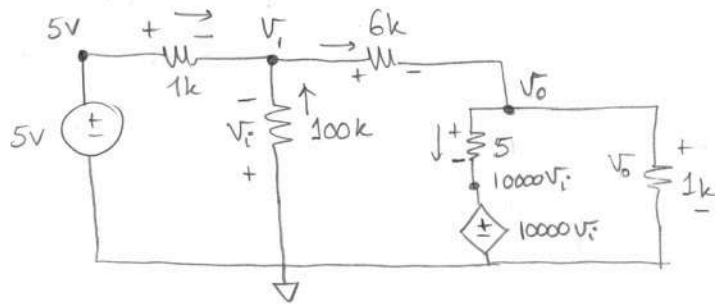
KVL @ supermesh: $-5000(i_1 - i_3) - 3 + 1000 i_2 + 5 = 0$

Constraint eq.: $i_2 - i_3 = 100 i_b$, where $i_b = i_1 - i_2 \Rightarrow i_2 - i_3 = 100(i_1 - i_2)$

Now simplify

$$\left. \begin{aligned} 7000 i_1 - 5000 i_3 &= -3 \\ -5000 i_1 + 1000 i_2 + 5000 i_3 &= -2 \\ -100 i_1 + 101 i_2 - i_3 &= 0 \end{aligned} \right\}$$

• QUESTION 2b



i)

$$V_i = -V_i$$

$$\text{KCL@ } V_i: \frac{5-V_i}{1k} + \frac{0-V_i}{100k} - \frac{V_i-V_o}{6k} = 0 \quad (1)$$

$$\text{KCL@ } V_o: \frac{V_i-V_o}{6000} - \frac{V_o-10000V_i}{5} - \frac{V_o}{1000} = 0 \quad (2)$$

Substitute $V_i = -V_i$ and solve for V_o and V_i

$$(1) \quad 3000 - 600V_i - 6V_i - 100V_i + 100V_o = 0$$

$$(2) \quad V_i - V_o - 1200V_o + 12000000(-V_i) - 6V_o = 0$$

$$\begin{aligned} -706V_i + 100V_o &= -3000 & \left(\begin{array}{l} (\times 1207) \\ (\times 100) \end{array} \right. & \begin{array}{l} -852142V_i + 120700V_o = 3621000 \\ -11999999V_i - 1207V_o = 0 \end{array} \end{aligned}$$

$$-1200852042V_i = 3621000$$

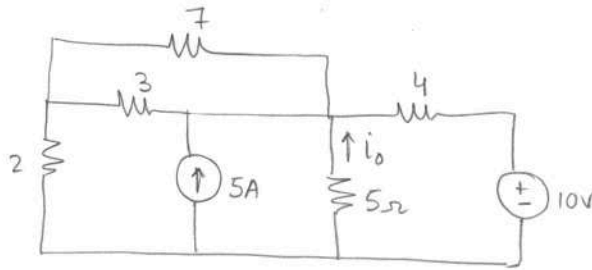
$$V_i = -0.00302V$$

$$-706V_i + 100V_o = -3000 \Rightarrow \boxed{V_o = \frac{-3000 + 706V_i}{100} = -30.02V}$$

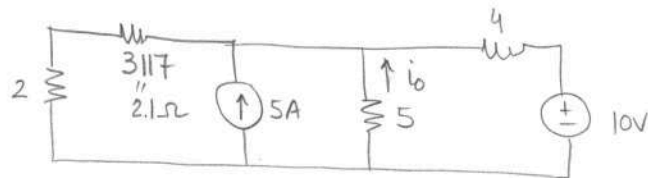
ii)

$$\boxed{\frac{V_o}{V_s} = \frac{-30.02}{5V} = -6}$$

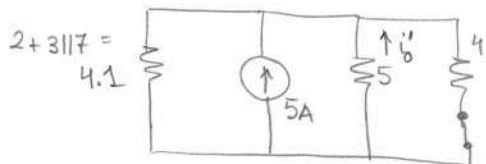
• QUESTION 3a



↓



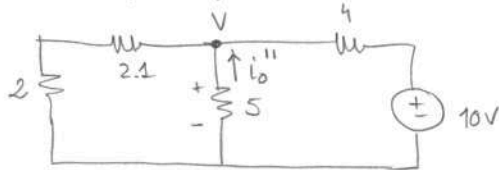
* Contribution of current source



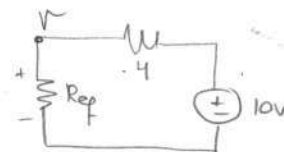
$$i_o' = - \frac{4.1 \parallel 5 \parallel 4}{8} = -1.441 \text{ A}$$

current divider

* Contribution of voltage source



⇒



$$R_{eq} = 5 \parallel (2.1 + 2) = 2.253$$

$$V = 10 \frac{R_{eq}}{4 + R_{eq}} = 10 \times \frac{2.253}{6.253} = 3.6 \text{ V}$$

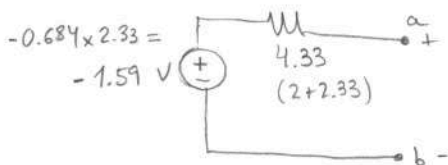
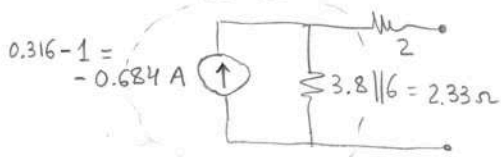
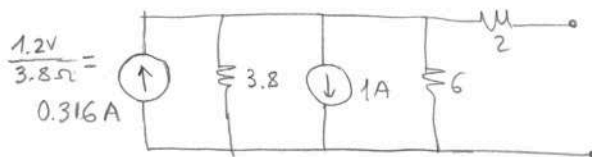
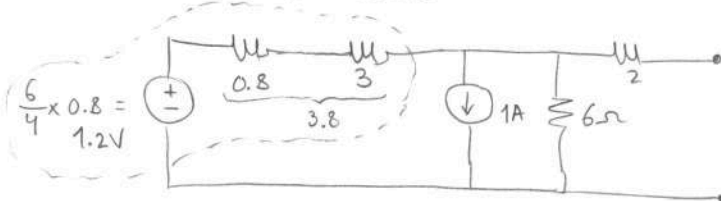
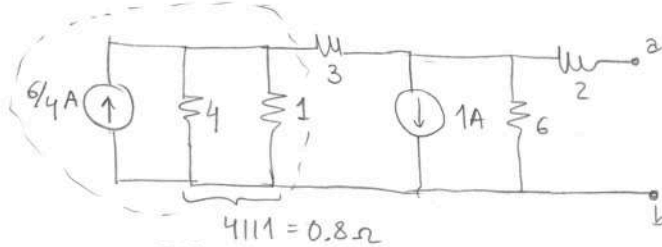
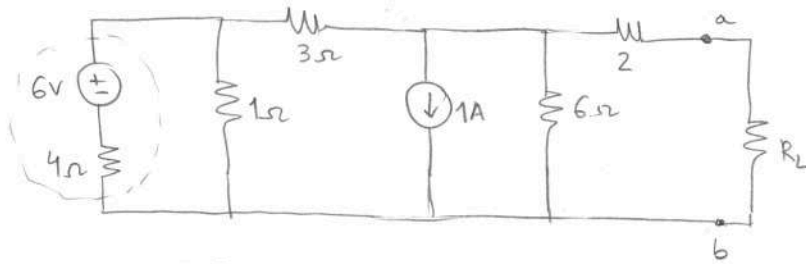
voltage divider

$$i_o'' = -\frac{V}{5} = -\frac{3.6}{5} = -0.72 \text{ A}$$

* Add contributions

$$i_o = i_o' + i_o'' = -1.441 - 0.72 = -2.161 \text{ A}$$

• QUESTION 3b

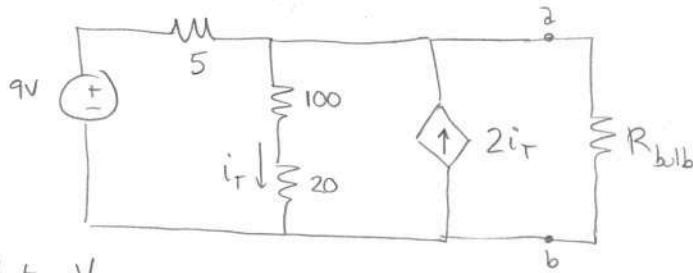


$$V_{Th} = V_{ab} = -1.59 \text{ V}$$

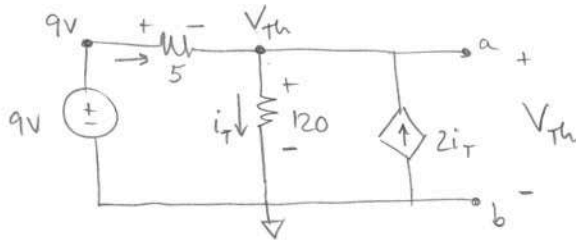
$$R_{Th} = 4.33 \Omega$$

Note: $V_{Th} = 1.59 \text{ V}$ with reversed polarity in the source is equivalent and also correct.

QUESTION 4



i) Calculate V_{Th} :



Using nodal analysis:

KCL @ node V_{Th} :

$$\frac{9 - V_{Th}}{5} - \frac{V_{Th}}{120} + 2i_T = 0, \text{ where } i_T = \frac{V_{Th}}{120}$$

$$\frac{9 - V_{Th}}{5} - \frac{V_{Th}}{120} + \frac{V_{Th}}{60} = 0 \quad \times 120$$

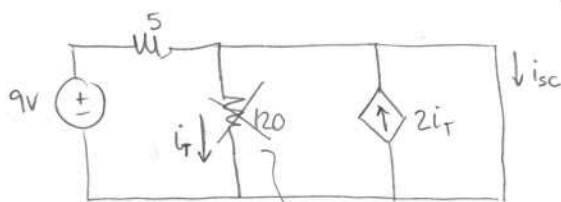
$$216 - 24V_{Th} - V_{Th} + 2V_{Th} = 0$$

$$216 - 23V_{Th} = 0 \Rightarrow \boxed{V_{Th} = \frac{-216}{-23} = 9.39V}$$

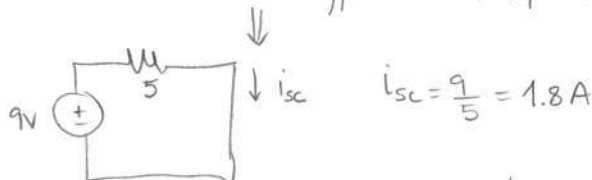
ii) Calculate R_{Th} (method 2)



$$R_{Th} = \frac{V_{Th}}{i_{sc}} \text{ and } R_L = R_{Th}$$



bypassed $\Rightarrow i_T = 0A \Rightarrow$ dependent source is open circuit



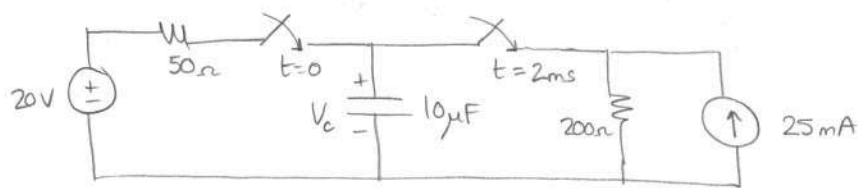
$$i_{sc} = \frac{9}{5} = 1.8A$$

$$R_L = R_{Th} = \frac{V_{Th}}{i_{sc}} = \frac{9.39}{1.8} = 5.22 \Omega$$

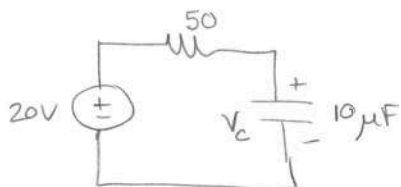
iii)

$$\boxed{P_{max} = \frac{V_{Th}^2}{4R_{Th}} = \frac{9.39^2}{4 \times 5.22} = 4.223W}$$

QUESTION 5



i) $0 \leq t < 2\text{ms}$ (switch 1 closed, switch 2 open)



Step response

$$V_c(t) = V_c(\infty) + [V_c(0) - V_c(\infty)] e^{-t/\tau} \text{ where } \tau = RC$$

* Calculate $V_c(0) \Rightarrow V_c(0) = 0\text{V}$

* Calculate $V_c(\infty)$:

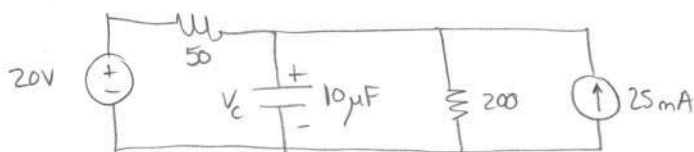


$$V_c(\infty) = 20\text{V}$$

* Calculate $\tau = RC = 50 \times 10\mu\text{F} = 500\mu\text{s} = 0.5\text{ms}$

* Substitute:
$$V_c(t) = 20 + [0 - 20] e^{-t/0.5 \times 10^{-3}} = 20(1 - e^{-2000t}) \text{ V}$$

ii) $t \geq 2\text{ms}$ (both switches closed)



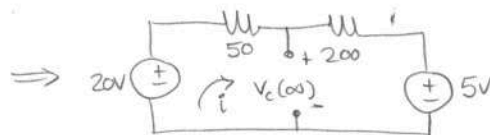
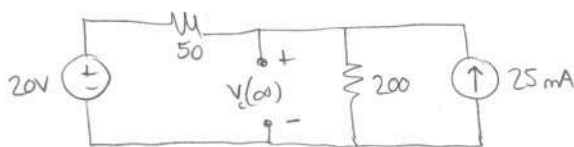
Step response

$$V_c(t) = V_c(\infty) + [V_c(t_0) - V_c(\infty)] e^{-(t-t_0)/\tau} \text{ where } \tau = RC$$

* Calculate $V_c(0.002)$ from previous interval

$$V_c(0.002) = 20(1 - e^{-2000 \times 0.002}) = 20(1 - 0.0183) = 19.634\text{V}$$

* Calculate $V_c(\infty)$



$$\text{KVL: } -20 + 50i + 200i + 5 = 0$$

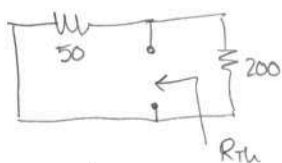
$$250i = 15 \Rightarrow i = \frac{15}{250} = 0.06\text{A}$$

$$\text{KVL @ open mesh: } -20 + 50i + V_c(\infty) = 0$$

$$V_c(\infty) = 20 - 50i = 20 - 50 \times 0.06 = 17\text{V}$$

* Calculate $\tau = R_{Th} \times C$

→ Calculate R_{Th} (method 1):



$$R_{Th} = 50 \parallel 200 = 40\Omega$$

$$\tau = R_{Th} \times C = 40 \times 10\mu\text{F} = 400\mu\text{s} = 0.4\text{ms}$$

* Substitute:
$$V_c(t) = 17 + [19.634 - 17] e^{-(t-0.002)/0.0004} = 17 + 2.634 e^{-2500(t-0.002)} \text{ V}$$